5G RAN

Virtual Grid-based Multi-Frequency Coordination Feature Parameter Description

Issue Draft A

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for Virtual Grid-based Multi-Frequency Coordination by the UMPThb. For details, see **4.3.3 Hardware**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Optimized the policy of collaboration between Virtual Grid-based Multi-Frequency Coordination and RF channel intelligent shutdown. For details, see: • **4.2.2 Impacts** • **4.3.2.3 Function Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **VIRTUAL_GRID_CH_SD_COOP_OPT_SW** option to the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between Virtual Grid-based Multi-Frequency Coordination and RF module fast deep dormancy. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Replaced the reserved option that controls "optimization of the policy of collaboration between Virtual Grid-based Multi-Frequency Coordination and TTI-level channel shutdown as well as dynamic RF channel shutdown" with an option of a formal parameter. For details, see: • **4.3.2.3 Function Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Replaced **RSVDSWPARAM4_BIT16** of **gNBRsvd**.***RsvdSwParam4*** with **VIRTUAL_GRID_CH_SD_COEXIST_SW** of **gNBPwrSavingAlgo**.***PwrSavingAlgoSwitch***. • Modified parameter: Added the **VIRTUAL_GRID_CH_SD_COEXIST_SW** option to the **gNBPwrSavingAlgo**.***PwrSavingAlgoSwitch*** parameter. • Added the following reserved option to the disuse list: **RSVDSWPARAM4_BIT16** of the **gNBRsvd**.***RsvdSwParam4*** parameter	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an activation verification method for Virtual Grid-based Multi-Frequency Coordination. For details, see **4.4.2 Activation Verification**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Reorganized this document, mainly involving rewording of information.

  – The editorial changes that involve changes in meanings are as follows:
    - Revised descriptions of virtual grid models and added a figure illustrating virtual grids. For details, see **3 Overview**.
    - Revised descriptions of the virtual grid model building process. For details, see **4.1.1 Virtual Grid Model Building**.
    - Revised descriptions of online model switching for different types of virtual grid models. For details, see **4.1.1 Virtual Grid Model Building**.
    - Revised descriptions of how virtual grid models are used in different functions. For details, see **4.1.2 Use of Virtual Grid Models**.
    - Revised descriptions of model updating, stop of model building and use, and model deletion. For details, see **4.1.3 Virtual Grid Model Maintenance**.

  – Other editorial changes do not involve changes in meanings and therefore are not detailed. Examples of such changes include chapter/section reorganization and text-to-table transformation.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-060201	Virtual Grid-based Multi-Frequency Coordination	4 Virtual Grid-based Multi-Frequency Coordination
FBFD-091150	Spectral Efficiency Grid	4 Virtual Grid-based Multi-Frequency Coordination

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Virtual Grid-based Multi-Frequency Coordination	None	4 Virtual Grid-based Multi-Frequency Coordination

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Virtual Grid-based Multi-Frequency Coordination	Supported only in low frequency bands	4 Virtual Grid-based Multi-Frequency Coordination

3 Overview

Background

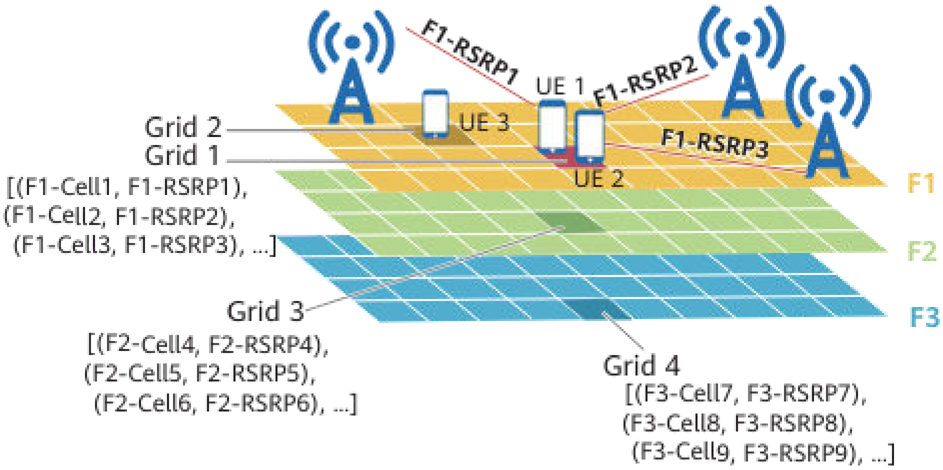
On multi-frequency networks, it is usually necessary for base stations to perform inter-frequency handovers or carrier selection for UEs in order to provide an optimal user experience.

- Inter-frequency handovers depend on inter-frequency measurements. However, inter-frequency measurements decrease UE throughput, especially for cell edge users (CEUs). In addition, due to inter-frequency measurement delay, optimal carriers cannot be promptly selected for UEs. Against this backdrop, virtual grid models are introduced to predict inter-frequency reference signal received power (RSRP), thus eliminating the need for inter-frequency measurements and improving user experience.
- A gNodeB selects carriers for a UE based on the RSRP reported by the UE (the serving carriers of the UE may be changed after the selection). However, the selected carriers may not be able to provide an optimal user experience. The virtual grid technology can predict the spectral efficiency of individual UEs on different carriers and can therefore be used to select carriers that can provide an optimal user experience.

Virtual Grid Model

A virtual grid is a signal grid of a cell. UEs in the same area of a cell normally have the same radio signal characteristics, including the reference signal received power (RSRP) and other information of the serving cell and intra-frequency neighboring cells. Based on the measurements of radio signals, UEs with the same radio characteristics are categorized as a single group. The gNodeB considers these UEs to be located in one signal grid, which is referred to as a virtual grid. Virtual grids are defined by UEs' signal characteristics instead of geographical locations., as shown in Figure 3-1. For example, if the measurement result for a frequency measured by UE 1 is [(F1-Cell 1, F1-RSRP 1), (F1-Cell 2, F1-RSRP 2), (F1-Cell 3, F1-RSRP 3), ...] and the measurement result for the same frequency measured by UE 2 is the same as that of UE 1, UE 1 and UE 2 are considered to be in the same virtual grid (grid 1) even if UE 1 and UE 2 are not in the same geographical location.

Figure 3-1 Virtual grids



The gNodeB uses virtual grids as key features to construct a signal characteristics mapping between a frequency and all virtual grids in a cell. This mapping is contained in an entity called a virtual grid model. The specific types of virtual grid models are listed in **Table 3-1**.

Table 3-1 Virtual grid models

Model Type	Model Principles
RSRP prediction model	After the gNodeB obtains the virtual grid information about a UE on a frequency, the gNodeB queries the RSRP prediction model for a neighboring frequency to quickly predict the RSRP of the UE on this neighboring frequency. This model is built for a cell served by a gNodeB on a per neighboring frequency basis.
Downlink intra-frequency spectral efficiency prediction model	After the gNodeB obtains the virtual grid information about a UE on a frequency, the gNodeB queries the downlink spectral efficiency model for this frequency to quickly predict the spectral efficiency of the UE on this frequency. This model is built separately for each cell served by a gNodeB.
Downlink-frequency spectral efficiency model	After the gNodeB obtains the virtual grid information about a UE on a frequency, the gNodeB queries the downlink spectral efficiency model for this frequency to quickly predict the spectral efficiency of the UE on this frequency. This model is built jointly for multiple cells operating on a specific frequency of a gNodeB.
Intra-frequency sounding reference signal (SRS) beam prediction model	After the gNodeB obtains the virtual grid information about a UE on a frequency, the gNodeB queries the intra-frequency SRS beam prediction model for this frequency to quickly predict the ID of the SRS beam with the largest SRS RSRP on this frequency. This model is built separately for each cell served by a gNodeB.

Virtual grid models described in this document refer only to intra-NR virtual grid models. For the descriptions of inter-RAT virtual grid models involving NR and LTE, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

4 Virtual Grid-based Multi-Frequency Coordination

4.1 Principles

4.1.1 Virtual Grid Model Building

Starting Model Building

A virtual grid model can be built after the corresponding function switches are turned on. **Table 4-1** lists the function switches on the gNodeB that control the building of different types of virtual grid models.

Table 4-1 Switches that control virtual grid model building

Model Type	Function Switch
RSRP prediction model	**VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter and **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter
Downlink intra-frequency spectral efficiency prediction model	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter and **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter
Downlink-frequency spectral efficiency model	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter
Intra-frequency SRS beam prediction model	**INTRAF_SRS_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter

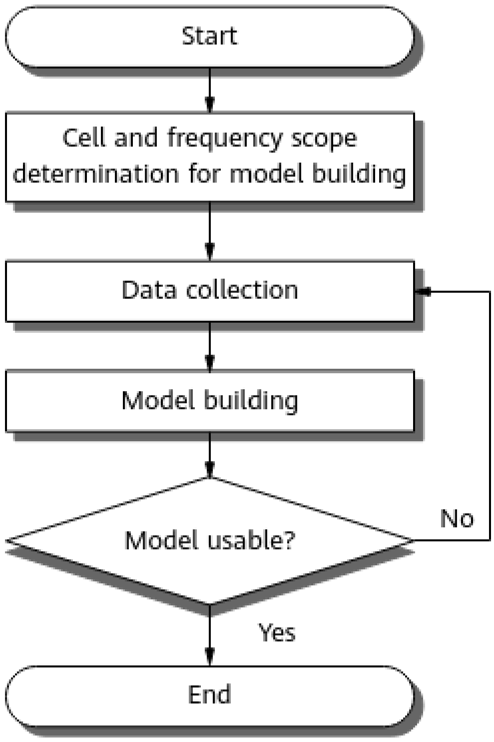
Model Building Process

There are two modes of virtual grid model building: fast building and weekly building.

- Fast building

  Figure 4-1 shows the process of fast virtual grid model building.

  Figure 4-1 Process of fast virtual grid model building

  a. **The gNodeB determines the scope of cells and frequencies for which virtual grid models are to be built.**

     - In the case of RSRP prediction models, the scope is determined based on the attributes of the cells and neighboring frequencies.
     - In the case of downlink intra-frequency spectral efficiency prediction models, the scope is determined based on the cell attributes.
     - In the case of downlink-frequency spectral efficiency models, the scope is determined based on the attributes of the cells and their operating frequencies.
     - In the case of intra-frequency SRS beam prediction models, the scope is determined based on the cell attributes.

     When new frequencies are configured and the number of virtual grid models has not reached the maximum, the gNodeB uses the configuration information of these frequencies to determine the frequencies for which models can be built. It then builds models for the determined frequencies according to the frequency activation sequence. For RSRP prediction models, when the **NR_TO_NR_COV_GRID_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected, model building is simplified and the maximum allowed number of models is increased. For details about the maximum allowed number of models, see **4.3.3 Hardware**.

b. **In each cell for which the gNodeB will build a virtual grid model, the gNodeB randomly selects 15 UEs every 40 seconds for data collection.** The data collection is complete when the number of samples accumulated reaches the storage limit or when data has been collected for 24 hours. **The gNodeB collects different data to use as samples for training different types of models.**

- To train RSRP prediction models, it collects measurement reports from UEs on different frequencies.
- To train downlink intra-frequency spectral efficiency prediction models, it collects information on the PRB usage of the strongest intra-frequency neighboring cell, the UE reception capability, and the spectral efficiency of the local cell.
- To train downlink-frequency spectral efficiency models, it collects information on the PRB usage of all intra-frequency neighboring cells, the UE reception capability, and the spectral efficiency of the local cell.
- To train intra-frequency SRS beam prediction models, it collects information on the UE transmit capability and the SRS beams of the local cell.

During data collection, the gNodeB normally delivers measurement configurations to a UE through an RRC Reconfiguration message. The UE performs measurement based on the measurement configurations and sends measurement reports to the gNodeB. Measurements are triggered periodically or based on events, depending on the setting of the **NR_VG_MDL_MR_IMP_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter.

- If this option is deselected, the gNodeB triggers measurements periodically.
- If this option is selected, the gNodeB triggers measurements based on events.
  - Event A1 configurations are used for measurements of the serving cell. Event A1 indicates that the signal quality of the serving cell becomes higher than a specific threshold. The threshold has a fixed value of –156 dBm.
  - Event A3 configurations are used for measurements of intra-frequency neighboring cells. Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset. The offset has a fixed value of –30 dB.
  - Event A5 configurations are used for measurements of inter-frequency neighboring cells. Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and that the signal quality of a neighboring cell exceeds threshold 2.

Threshold 1 has a fixed value of –31 dBm and threshold 2 has a fixed value of –156 dBm.

c. **The gNodeB starts model training and calculates the model accuracy.**
   - In the case of RSRP prediction models:
     When the **NR_COV_VG_MDL_WITH_TA_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is selected, timing advance (TA) information is added during the building and use of grid models to improve the model accuracy.
   - In the case of downlink intra-frequency spectral efficiency prediction models:
     - When the **INTRAF_SPCT_EFF_PRED_ACCU_IMP** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is selected, the serving cell can calculate the actual downlink spectral efficiency of UEs and then use the result to replace the spectral efficiency predicted by grid models, thereby improving the model accuracy.
     - When the **DL_INTRA_EFF_MDL_MULTI_SSB_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is selected, "association of downlink intra-frequency spectral efficiency grid models with the characteristics of multiple SSB beams" is enabled. This function allows information about multiple SSB beams to be used during model building and model use, thereby improving the model accuracy.

d. **The gNodeB evaluates whether each model can be used based on the model's accuracy.** A model will fail to be built if the sample data collected is insufficient or the prediction accuracy does not meet the application requirements. In this case, the process goes back to **2**. If a model fails to be built seven consecutive times, it cannot be built again for 28 days.

- Weekly building

  After a model is successfully built for a cell, the gNodeB randomly selects three UEs in the cell every 40 seconds. It collects sample data from the three UEs for seven days to build another model. The building process is the same as that described in **Fast building** (only the data collection time is different). After the model is successfully built, the gNodeB compares the model's accuracy with that of the existing model. If the accuracy of the weekly built model is higher than that of the existing model, the weekly built model goes online and the existing model goes offline. Otherwise, the existing model is kept in use.

NOTE

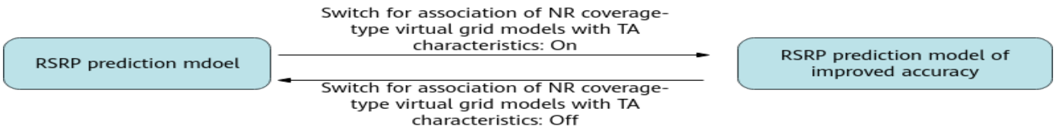
If a cell is deactivated while a virtual grid model is being built for the cell, the model building process will be suspended and the collected samples will be saved. After the cell is reactivated, data collection and model building continue. If the cell remains deactivated for a long time, the model accuracy may be low, or the model building even fails. In case the model building fails, a model can be built at the next building attempt.

Model Switching

Certain types of virtual grid models involve switching between models of the same type but different levels of accuracy or between different models, and this leads to a change of the models in use. Intra-frequency SRS beam prediction models do not involve model switching. Models that involve model switching are as follows:

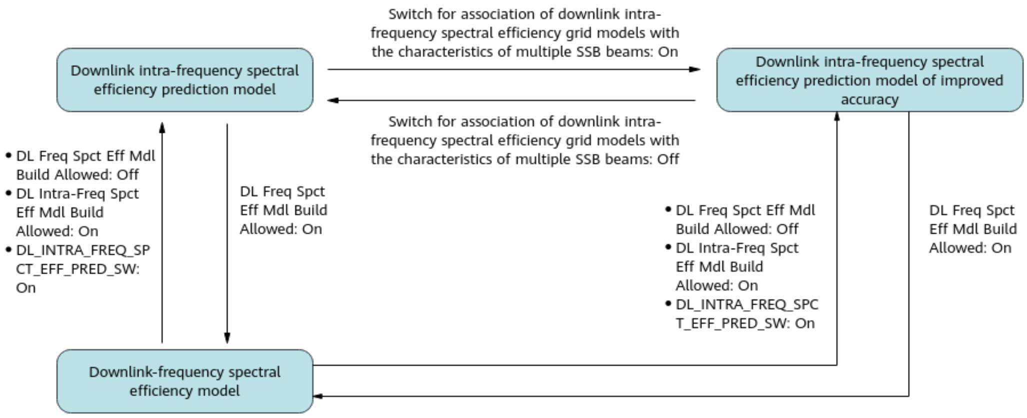
- RSRP prediction models: There can be switching between RSRP prediction models of different levels of accuracy, as shown in **Figure 4-2**.

  Figure 4-2 Switching between RSRP prediction models of different levels of accuracy



- Downlink intra-frequency spectral efficiency prediction models and downlink-frequency spectral efficiency models: There can be switching between downlink intra-frequency spectral efficiency prediction models of different levels of accuracy, and switching between a downlink intra-frequency spectral efficiency prediction model (regardless of the accuracy) and a downlink-frequency spectral efficiency model, as shown in **Figure 4-3**.

  Figure 4-3 Switching between downlink intra-frequency spectral efficiency prediction models and between a downlink intra-frequency spectral efficiency prediction model and a downlink-frequency spectral efficiency model



Specifically:

- In switching between models of the same type but different levels of accuracy, the old model goes offline immediately, and then a new model is built and comes online. The features that used the old model can no longer use the old model. After the new model comes online, the features that used the old model start to use the new model.
- In switching between a downlink intra-frequency spectral efficiency prediction model (regardless of the accuracy) and a downlink-frequency spectral efficiency model:
  - If the switching is from a downlink intra-frequency spectral efficiency prediction model to a downlink-frequency spectral efficiency model, the downlink-frequency spectral efficiency model is built and comes online first, and then the downlink intra-frequency spectral efficiency prediction model goes offline. Features that used the downlink intra-frequency spectral efficiency prediction model start to use the downlink-frequency spectral efficiency model.

– If the switching is from a downlink-frequency spectral efficiency model to a downlink intra-frequency spectral efficiency prediction model, the downlink-frequency spectral efficiency model goes offline immediately, and then the downlink intra-frequency spectral efficiency prediction model is built and comes online. Features that used the downlink-frequency spectral efficiency model can no longer use the model. After the downlink intra-frequency spectral efficiency prediction model comes online, features that used the downlink-frequency spectral efficiency model start to use the downlink intra-frequency spectral efficiency prediction model.

4.1.2 Use of Virtual Grid Models

Successfully-built virtual grid models need to be used in different functions to improve user experience

4.1.2.1 Use of RSRP Prediction Models

After an RSRP prediction model is successfully built, whether the model can be used in the corresponding cell depends on the value of the **NRCellSmartMultiCarr**.***CovGridModelAppPolicy*** parameter.

- If this parameter is set to **FAST_MODE**, an RSRP prediction model can be used as long as it is successfully built for any neighboring frequency.
- If this parameter is set to **STRICT_MODE**, RSRP prediction models can be used only after they are successfully built for all neighboring frequencies.

The prediction results provided by RSRP prediction models can be used for intra-NR inter-frequency handovers, carrier selection, and SCell addition.

Intra-NR Inter-Frequency Handovers

When the inter-frequency handover functions listed in **Table 4-2** are enabled and the corresponding switches controlling the use of virtual grids are turned on, the gNodeB uses the prediction results provided by RSRP prediction models in corresponding inter-frequency handovers. For details about inter-frequency handover functions, see *SA Mobility Management in Connected Mode*.

Table 4-2 Switches related to the use of RSRP prediction models in inter-frequency handovers

Inter-Frequency Handover Function	Function Switch	Switch Controlling the Use of Virtual Grids
Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	**COV_IF_HO_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***AiInterFreqHoOptSwitch*** parameter

Inter-Frequency Handover Function	Function Switch	Switch Controlling the Use of Virtual Grids
Service-based inter-frequency handover	**SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	**SERV_IF_HO_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***AiInterFreqHoOptSwitch*** parameter
Uplink-interference-based inter-frequency handover	**UL_INTRF_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	**ULINTRF_IF_HO_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***AiInterFreqHoOptSwitch*** parameter
Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch.*InterFreqHoSwitch*** parameter	**FRQ_PRI_IF_HO_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***AiInterFreqHoOptSwitch*** parameter

In an intra-NR inter-frequency handover, the gNodeB delivers event A3 measurement configurations related to the serving frequency to the UE. After receiving an event A3 measurement report from the UE, the gNodeB uses RSRP prediction models to query the RSRP values of candidate neighboring frequencies based on the report. The candidate neighboring frequencies are those left after frequency filtering.

- Assume that the cell serving the UE has an RSRP prediction model for a candidate neighboring frequency and the model's accuracy meets the requirement of inter-frequency handover algorithms. Whether the gNodeB can use the RSRP prediction model to predict the RSRP of this candidate neighboring frequency depends on the enabling status of the grid-based handover protection function (controlled by the **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr**.***HoProtectAlgoSwitch*** parameter).
  - If grid-based handover protection is disabled, the gNodeB can use the RSRP prediction model to predict the RSRP of the neighboring frequency and allow a grid-based handover to a cell on the neighboring frequency. The prediction eliminates the need for independent measurements on the neighboring frequency.
  - If grid-based handover protection is enabled, the gNodeB prohibits grid-based handovers to cells on the neighboring frequency. In this case, inter-frequency or inter-RAT measurements are still required for points that are predicted to have coverage. The gNodeB will deliver event A4/A5 measurement configurations to the UE and start a timer at the same time. The measurements will be stopped once the gNodeB receives measurement reports or the timer expires.
- If the cell serving the UE does not have an RSRP prediction model for any candidate neighboring frequency or if no RSRP prediction model meets the accuracy requirement of inter-frequency handover algorithms, the gNodeB will deliver event A4/A5 measurement configurations to the UE and start a

timer at the same time. The measurements will be stopped once the gNodeB receives corresponding measurement reports or the timer expires.

The prediction accuracy required by inter-frequency handover algorithms is specified by the **NRCellSmartMultiCarr**.***PredAccuForDataThld*** and **NRCellSmartMultiCarr**.***PredAccuForVoiceThld*** parameters. For details about frequency filtering, see *SA Mobility Management in Connected Mode*.

The process of using RSRP prediction models' prediction results in inter-frequency handovers is as follows:

1. The gNodeB determines the range of inter-frequency neighboring cells for selection.
2. The gNodeB filters inter-frequency neighboring cells based on RSRP.
3. The gNodeB sorts inter-frequency neighboring cells in descending order of the sum of the predicted RSRP value, *Ofn*, and *Ocn*.
4. The gNodeB initiates handover attempts in order based on the sort result.

The requirements of each inter-frequency handover function for inter-frequency neighboring cells are listed in **Table 4-3**.

Table 4-3 Requirements of each inter-frequency handover function for inter-frequency neighboring cells

Inter-Frequency Handover Function	Inter-Frequency Measurement Event	Range of Inter-Frequency Neighboring Cells for Selection	RSRP Requirement for Inter-Frequency Neighboring Cells
Coverage-based inter-frequency handover	A5	All neighboring cells with predicted RSRP values on each frequency	Predicted RSRP value + *Ofn* + *Ocn* – *Hys* > **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2*** + **gNBOperatorQciParam**.***InterFreqA5RsrpThld2Ofs***
Coverage-based inter-frequency handover	A3	All neighboring cells with predicted RSRP values on each frequency	Predicted RSRP value + *Ofn* + *Ocn* – *Hys* > Measured RSRP value of the serving cell + **NRCellInterFHoMeaGrp**.***InterFreqHoA3Offset***

Inter-Frequency Handover Function	Inter-Frequency Measurement Event	Range of Inter-Frequency Neighboring Cells for Selection	RSRP Requirement for Inter-Frequency Neighboring Cells
Service-based inter-frequency handover	A4/A5	Strongest neighboring cell on each frequency as indicated by the sum of the predicted RSRP value and *Ocn*	Predicted RSRP value + *Ofn* + *Ocn* – *Hys* > **NRCellInterFHoMeaGrp**.***SrvInterFreqA4RsrpThld***
Uplink-interference-based inter-frequency handover	A3	Strongest neighboring cell on each frequency as indicated by the sum of the predicted RSRP value and *Ocn*	Both of the following conditions are met: ● Predicted RSRP value + *Ofn* + *Ocn* – *Hys* > Measured RSRP value of the serving cell + **NRCellInterFHoMeaGrp**.***IntrfInterFreqHoA3RsrpOfs*** ● Predicted RSRP value > **NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld***
Frequency-priority-based inter-frequency handover	A4/A5	Strongest neighboring cell on each frequency as indicated by the sum of the predicted RSRP value and *Ocn*	Predicted RSRP value + *Ofn* + *Ocn* – *Hys* > **NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld***

Note:

- *Ofn* is the frequency-specific offset for RRC_CONNECTED UEs and is specified by the **NRCellFreqRelation**.***ConnFreqOffset*** parameter.
- *Ocn* is the cell individual offset (CIO) and is specified by the **NRCellRelation**.***CellIndividualOffset*** parameter. If the value is not **DB0**, this parameter is included in measurement configurations. If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
- *Hys* is the hysteresis for events related to inter-frequency measurements and is specified by the **NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst*** parameter.

NOTE

If a frequency reported for a frequency-priority-based inter-frequency handover has the highest traffic priority specified by the **NRCellFreqRelation**.***TrafficPriority*** parameter, the frequency will be determined as the target frequency and the UE will be immediately handed over to the frequency.

Carrier Selection

Carrier selection refers to experience-based smart carrier selection, multi-frequency beam coordination, and latency-based optimal carrier selection. When the carrier selection functions listed in **Table 4-4** are enabled, the gNodeB uses the prediction results provided by RSRP prediction models in these functions.

Table 4-4 Use of RSRP prediction models

Function Name	Function Switch	Use of Models
Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	The gNodeB inputs the RSRP values of the serving cell and intra-frequency neighboring cells contained in the measurement reports into the RSRP prediction models to obtain inter-frequency RSRP values. The gNodeB then uses the information obtained to determine the neighboring frequencies for which event A5 measurement configurations will be delivered.
Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	When the **MF_BEAM_CO_EXT_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected, the gNodeB uses RSRP prediction models to predict inter-frequency RSRP values. Based on the predicted RSRP values, the gNodeB determines the neighboring frequencies for which event A5 measurement configurations will be delivered.
Latency-based optimal carrier selection	**DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	The gNodeB inputs the RSRP values of the serving cell and intra-frequency neighboring cells contained in the measurement reports into the RSRP prediction models to obtain inter-frequency RSRP values. The gNodeB then uses the information obtained to determine the neighboring frequencies for which event A5 measurement configurations will be delivered.

NOTE

For details about experience-based smart carrier selection, multi-frequency beam coordination, and latency-based optimal carrier selection, see *Multi-Frequency Smart Selection*.

SCell Configuration

SCell configuration refers to SCell configuration for CA and SUL carrier addition for super uplink. When the functions involving SCell configuration are enabled and the **CARR_ADD_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected, the gNodeB uses the prediction results provided by RSRP prediction models in SCell configuration, as described in **Table 4-5**.

Table 4-5 Use of RSRP prediction models

Function Name	Function Switch	Use of Models
Carrier aggregation	• Downlink intra-band CA: **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter • Downlink intra-FR inter-band CA: **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	• If the model-predicted RSRP of the strongest inter-frequency neighboring cell on a frequency is greater than or equal to the sum of **NRCellCaMgmtConfig**.***CaA5RsrpThld2*** and **gNBCaFrequency**.***CaSccA5RsrpThld2Offset***, the gNodeB considers this inter-frequency cell as a candidate SCell. • If the model-predicted RSRP of the strongest inter-frequency neighboring cell on a frequency is less than the sum of **NRCellCaMgmtConfig**.***CaA5RsrpThld2*** and **gNBCaFrequency**.***CaSccA5RsrpThld2Offset***, the gNodeB filters out the measurements of this frequency and checks the next neighboring frequency. • Finally, the gNodeB selects the cell with the highest RSRP among all candidate SCells and configures this cell as an SCell.

Function Name	Function Switch	Use of Models
Super uplink	**FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	• If the model-predicted RSRP of the strongest inter-frequency neighboring cell on a frequency is greater than or equal to **NRCellSulFrequency**.***SulInterFreqA5RsrpThld2***, the gNodeB considers this inter-frequency cell as a candidate cell for SUL carrier addition. • If the model-predicted RSRP of the strongest inter-frequency neighboring cell on a frequency is less than **NRCellSulFrequency**.***SulInterFreqA5RsrpThld2***, the gNodeB filters out the measurements of this frequency and checks the next neighboring frequency. • Finally, the gNodeB selects the cell with the highest RSRP among all candidate cells for SUL carrier addition and configures the carrier of this cell as an SUL carrier.

NOTE

- For details about CA, see *Carrier Aggregation*.
- For details about super uplink, see *Super Uplink*.

4.1.2.2 Use of Downlink Intra-Frequency Spectral Efficiency Prediction Models and Downlink-Frequency Spectral Efficiency Models

When the functions listed in **Table 4-6** are enabled, the gNodeB uses the prediction results provided by downlink intra-frequency spectral efficiency prediction models or downlink-frequency spectral efficiency models in these functions.

Table 4-6 Use of downlink intra-frequency spectral efficiency prediction models and downlink-frequency spectral efficiency models

Function Name	Function Switch	Use of Models
Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	The gNodeB can use the downlink spectral efficiency predicted by the downlink intra-frequency spectral efficiency prediction model or downlink frequency spectral efficiency model to evaluate the downlink air interface capability.
Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	The gNodeB can use the downlink spectral efficiency predicted by the downlink intra-frequency spectral efficiency prediction model or downlink frequency spectral efficiency model to evaluate the downlink air interface capability.
Latency-based optimal carrier selection	**DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	The gNodeB can use the downlink spectral efficiency predicted by the downlink intra-frequency spectral efficiency prediction model or downlink frequency spectral efficiency model to evaluate the air interface latency.
Macro-micro intra-frequency optimal carrier selection	**INTRA_FREQ_COORD_HO_OPT_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter	The gNodeB can use the downlink spectral efficiency predicted by the downlink intra-frequency spectral efficiency prediction model or downlink frequency spectral efficiency model to evaluate the downlink air interface capability.

NOTE

For details about experience-based smart carrier selection, multi-frequency beam coordination, macro-micro intra-frequency optimal carrier selection, and latency-based optimal carrier selection, see *Multi-Frequency Smart Selection*.

4.1.2.3 Use of Intra-Frequency SRS Beam Prediction Models

When multi-frequency beam coordination is enabled, the gNodeB uses the prediction results provided by intra-frequency SRS beam prediction models in multi-frequency beam coordination, as described in **Table 4-7**.

Table 4-7 Use of intra-frequency SRS beam prediction models

Function Name	Function Switch	Use of Models
Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	The gNodeB can use the intra-frequency SRS beam prediction model to predict whether an SRS beam is within the range of candidate beams and further determines whether to instruct the involved UE to perform an inter-frequency handover.

NOTE

For details about multi-frequency beam coordination, see *Multi-Frequency Smart Selection*.

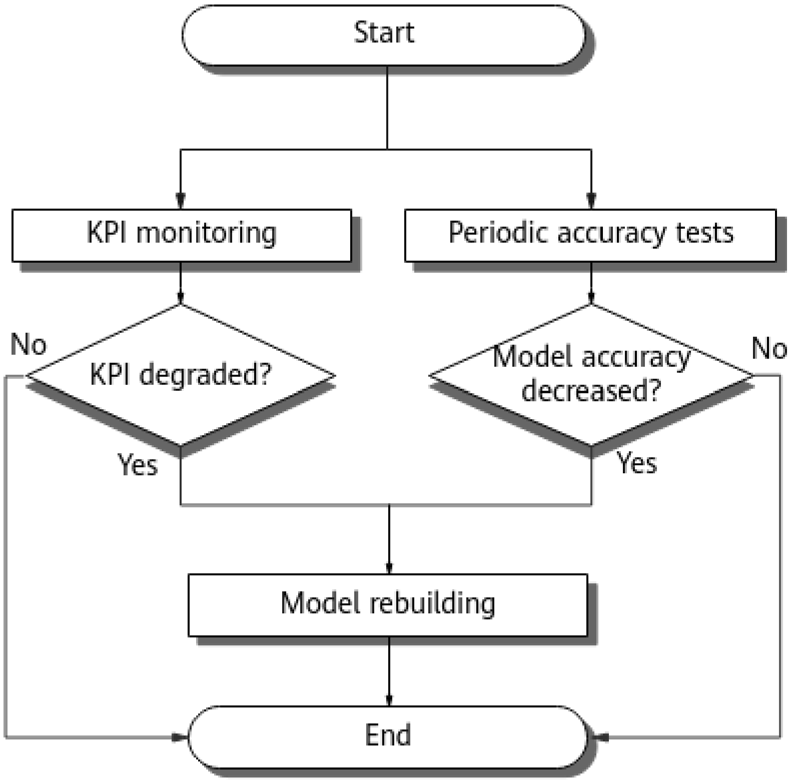
4.1.3 Virtual Grid Model Maintenance

Virtual grid model maintenance includes model updating, stop of model building and use, and model deletion.

Model Updating

After virtual grid models go online, the gNodeB monitors their performance and updates them. The following uses an RSRP prediction model as an example. **Figure 4-4** shows the process of monitoring the performance of the model and updating it.

Figure 4-4 Process of monitoring the performance of an RSRP prediction model and updating it



The gNodeB starts KPI monitoring and periodic accuracy tests on the model at the same time, that is, it simultaneously performs 1 and 2.

1. The gNodeB monitors the inter-frequency handover success rate in real time while the model is in use.

   If the success rate of inter-frequency handovers based on the model's prediction results (indicated by **Inter-Frequency Handover Out Success Rate (CU)**) is below the threshold specified by the **NRCellSmartMultiCarr**.***NrIfHoSuccRateProtectThld*** parameter for two consecutive hours, the gNodeB suspends the model. The model can be used again after it has been suspended for 22 hours.

   – If the inter-frequency handover success rate is below the preceding threshold in the first hour after the model is used again, the gNodeB deletes the model and goes to 3 to rebuild the model.
   – Otherwise, the gNodeB keeps the model in use.
2. The gNodeB performs a model accuracy test every third day.
   – If the accuracy of the model is lower than expected, the gNodeB considers the model to be aged. It then suspends the model, and goes to 3 to rebuild the model.
   – Otherwise, the gNodeB keeps the model in use.
3. The gNodeB selects UEs at random, collects data, and rebuilds the model. The new model will be used after it is successfully built.

Rebuilding of downlink intra-frequency spectral efficiency prediction models, intra-frequency SRS beam prediction models, and downlink-frequency spectral efficiency models can be triggered only by periodic accuracy tests. The accuracy tests are the same as those for RSRP prediction models.

Stop of Model Building and Use

When the switches listed in **Table 4-8** are turned off, RSRP prediction models and downlink intra-frequency spectral efficiency prediction models are no longer built or used. When the switch for building downlink-frequency spectral efficiency models or intra-frequency SRS beam prediction models is turned off, the corresponding models will be directly deleted. For details about model deletion, see **Model Deletion**.

Table 4-8 Stop of model building and use

Model Type	Switch Turned Off	Stop of Model Building and Use
RSRP prediction model	**NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMulti-Carr**.***NrMultiCarrierAlgoSwitch*** parameter	• The gNodeB will stop building and using the RSRP prediction models for the corresponding frequencies and will save the models that were successfully built. • The models can be saved for a maximum of seven days unless the gNodeB is reset. If this switch is turned on again within the seven days, the models will remain available.
Downlink intra-frequency spectral efficiency prediction model	**DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMulti-Carr**.***NrMultiCarrierAlgoSwitch*** parameter	• The gNodeB will stop building and using the downlink intra-frequency spectral efficiency prediction models for the cell and will save the models that were successfully built. • The models can be saved for a maximum of seven days unless the gNodeB is reset. If this switch is turned on again within the seven days, the models will remain available.

Model Deletion

When the switches listed in **Table 4-9** are turned off, the corresponding virtual grid models will be deleted. The number of deleted models will be excluded from the model count so that new models can be built.

Table 4-9 Model deletion

Model Type	Switch Turned Off	Model Deletion
RSRP prediction model	**VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter	The gNodeB will delete the RSRP prediction models for the corresponding frequencies so that new models can be built.
Downlink intra-frequency spectral efficiency prediction model	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter	The gNodeB will delete the downlink intra-frequency spectral efficiency prediction models for the cell so that new models can be built.
Downlink-frequency spectral efficiency model	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter	• The gNodeB will remove the cell from the cells engaged in the downlink-frequency spectral efficiency model so that another cell can be engaged in the model. • If this switch is turned off for all cells on a frequency, the downlink-frequency spectral efficiency model for the frequency will be deleted so that a new model can be built for another frequency.
Intra-frequency SRS beam prediction model	**INTRAF_SRS_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter	The gNodeB will delete the intra-frequency SRS beam prediction models for the cell so that new models can be built.

4.2 Network Analysis

4.2.1 Benefits

The Virtual Grid-based Multi-Frequency Coordination function allows inter-frequency measurements to be skipped, which can increase the **User Downlink Average Throughput (DU)**.

Before the Virtual Grid-based Multi-Frequency Coordination function is activated on a gNodeB, it is recommended that frequency configuration on the gNodeB be checked manually or using a related tool. If there are redundant or incorrect frequencies, delete or correct these frequencies. Otherwise, virtual grid models may fail to be built. It is recommended that the Virtual Grid-based Multi-Frequency Coordination function be enabled in the following scenarios:

- Multiple frequencies are used on a gNodeB.
- There are a large number of UEs. (You are advised to select cells where the value of **N.User.RRCConn.Avg** per hour is greater than 15. If the value is less than 2, it is possible that virtual grid models cannot be launched.)
- The traffic volume is stable. (You are advised to select cells where the **Cell Downlink Average Throughput (DU)** changes by no more than 10%.)
- Inter-frequency handovers are frequent.

NOTE

A downlink-frequency spectral efficiency model has a lower requirement for the number of UEs than a downlink intra-frequency spectral efficiency prediction model. A cell can be selected for downlink-frequency spectral efficiency model building as long as the value of **N.User.RRCConn.Avg** per hour is two or more in the cell. It is recommended that the downlink-frequency spectral efficiency model switch be turned on for all intra-frequency cells on a base station. It is required that the number of intra-base-station intra-frequency cells be greater than or equal to 3; otherwise, the grid model may fail to go online.

The greater the average number of inter-frequency handovers in cells served by a gNodeB and the greater the number of SCell configuration attempts before the Virtual Grid-based Multi-Frequency Coordination function is enabled, the greater the benefits. The numbers of inter-frequency handovers are measured by the **N.HO.InterFreq.Coverage.PrepAttOut**, **N.HO.InterFreq.Service.PrepAttOut**, and **N.HO.InterFreq.FreqPri.PrepAttOut** counters. The number of SCell configuration attempts is measured by the **N.CA.SCell.Add.Att** counter.

The benefits of the Virtual Grid-based Multi-Frequency Coordination function increase with the values of factors such as the proportion of UEs involved in coverage-, service-, uplink-interference-, and frequency-priority-based inter-frequency handovers, the proportion of CA UEs, the number of inter-frequency measurements, the loss caused by gap-assisted inter-frequency measurements, and the UE traffic volume.

Currently, the Virtual Grid-based Multi-Frequency Coordination function does not support UE handovers between base stations supplied by different vendors. This function is not recommended for base stations where such handovers are involved.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

Virtual Grid-based Multi-Frequency Coordination has the following impacts on the network:

- The gNodeB needs to select a certain number of UEs for virtual grid model building and updating. The selected UEs will perform gap-assisted inter-frequency measurements, which decrease the values of **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** for these UEs. Since only a small number of UEs are selected in a cell, the impacts on **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** can be ignored when there are more than 30 online UEs in the cell.
- Frequent virtual grid model queries increase the CPU usage (indicated by **Average CPU Load**) of the main control board.

- The building and use of virtual grid models increase the CPU usage. During model building and use, CPU usage is monitored based on the CPU flow control mechanism. If the CPU usage exceeds 60%, the models will temporarily enter the unavailable state. When the CPU usage decreases to less than 40%, the models will become available for use again.
- Intra- and inter-frequency measurements during virtual grid model building increase the number of signaling messages over the air interface. Therefore, signaling radio bearer (SRB) traffic increases. As SRBs use modulation and coding schemes (MCSs) with small indexes, the average MCS index may decrease. In addition, the transmission of RRC reconfiguration messages may fail, increasing the value of **Call Drop Rate (CU, VoNR)**.
- When Virtual Grid-based Multi-Frequency Coordination is enabled and grid-based handover protection (controlled by the **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr**.***HoProtectAlgoSwitch*** parameter) is disabled, the number of virtual-grid-based inter-frequency or inter-RAT handovers increases and the virtual-grid-based inter-frequency or inter-RAT handover success rate may decrease. Grid-based handover protection is enabled by default to reduce the impact on the inter-frequency or inter-RAT handover success rate.
- When Virtual Grid-based Multi-Frequency Coordination and RF channel intelligent shutdown are both enabled, optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions can be enabled by selecting the **VIRTUAL_GRID_CH_SD_COOP_OPT_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter to reduce the probability that building of virtual grid models is suspended and the probability that virtual grid models go offline and become unavailable due to RF channel intelligent shutdown.

  The impacts of changes in the setting of the option are as follows:
  - When the setting is changed from off to on, models incapable of collaboration optimization go offline and models capable of collaboration optimization come online. Building of the models capable of collaboration optimization requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models incapable of collaboration optimization go offline, features that used the models can no longer use them.
  - When the setting is changed from on to off, models capable of collaboration optimization go offline and models incapable of collaboration optimization come online. Building of the models incapable of collaboration optimization requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models capable of collaboration optimization go offline, features that used the models can no longer use them.
- When the prediction results of virtual grid models are used for handovers, the values of some counters related to the number of handovers may fluctuate. The probability of inter-frequency handover failures may increase for any of the following reasons, and the value of **Inter-Frequency Handover Out Success Rate (CU)** may decrease consequently:

- The blind handover performance of some UEs is poor. This increases the probability of inter-frequency handover failures when blind handovers are triggered based on the information predicted by virtual grid models.
- The network structure is changed due to RF optimization or other operations. When there is such a change, the gNodeB monitors and updates virtual grid models. During this process, the probability of inter-frequency handover failures increases.
- The predicted RSRP values in some grids are inaccurate. This increases the probability of inter-frequency handover failures at these grids.
- The predicted SSB IDs in some grids are inaccurate. This increases the probability of contention-based access for UEs in these grids and consequently increases the probability of inter-frequency handover failures.
- The status information of neighboring cells cannot be obtained due to an Xn link fault or flow control over the Xn link. If this happens, the prediction accuracy of grids may decrease, resulting in an increase in the probability of inter-frequency handover failures.

- Impacts specifically related to RSRP prediction models:
  - The change of the setting of the **NR_COV_VG_MDL_WITH_TA_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter has the following impacts on the network:
    - When the setting is changed from off to on, models without accuracy improvement go offline and models with accuracy improvement come online. Building of the models with accuracy improvement requires certain data from UEs, and the UEs involved in data collection will experience a decrease in their uplink and downlink throughputs. In addition, after the models without accuracy improvement go offline, features that used the models can no longer use them.
    - When the setting is changed from on to off, models with accuracy improvement go offline and models without accuracy improvement come online. Building of the models without accuracy improvement requires certain data from UEs, and the UEs involved in data collection will experience a decrease in their uplink and downlink throughputs. In addition, after the models with accuracy improvement go offline, features that used the models can no longer use them.
  - When the setting of the **NR_TO_NR_COV_GRID_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is changed from on to off, all existing models will go offline and will be rebuilt. When the setting of this option is changed from off to on, the existing models will not go offline and will not be rebuilt.
- Impacts specifically related to downlink intra-frequency spectral efficiency prediction models:
  - When the **INTRAF_SPCT_EFF_PRED_ACCU_IMP** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is selected, the downlink intra-frequency spectral efficiency prediction accuracy increases, but the CPU load also increases.

- The change of the setting of the **DL_INTRA_EFF_MDL_MULTI_SSB_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter has the following impacts on the network:
  - When the setting is changed from off to on, models without accuracy improvement go offline and models with accuracy improvement come online. Building of the models with accuracy improvement requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models without accuracy improvement go offline, features that used the models can no longer use them.
  - When the setting is changed from on to off, models with accuracy improvement go offline and models without accuracy improvement come online. Building of the models without accuracy improvement requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models with accuracy improvement go offline, features that used the models can no longer use them.

- Impacts specifically related to downlink-frequency spectral efficiency models:
  - If the **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected for a cell with a downlink intra-frequency spectral efficiency prediction model in use, the corresponding downlink-frequency spectral efficiency model will be built and come online, and then the downlink intra-frequency spectral efficiency prediction model goes offline. Building of the downlink-frequency spectral efficiency model requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads.
  - When the **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is deselected for a cell with a downlink-frequency spectral efficiency model in use and the switches for the downlink intra-frequency spectral efficiency prediction model are turned on, the downlink-frequency spectral efficiency model goes offline and the downlink intra-frequency spectral efficiency prediction model comes online. After the downlink-frequency spectral efficiency model goes offline, features that used the model can no longer use it. In addition, building of the downlink intra-frequency spectral efficiency prediction model requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. The switches for the downlink intra-frequency spectral efficiency prediction model are the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter and the **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter.

4.3 Requirements

4.3.1 Licenses

- RSRP prediction models, intra-frequency SRS beam prediction models, and downlink intra-frequency spectral efficiency prediction models require the following license.

RAT	Feature ID	Feature Name	Model	Sales Unit
Low-frequency TDD	FOFD-060201	Virtual Grid-based Multi-Frequency Coordination	NR0S00VGMC00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

- Downlink-frequency spectral efficiency models do not have license requirements.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based inter-frequency handover	**SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.
Low-frequency TDD	Uplink-interference-based inter-frequency handover	**UL_INTRF_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.
Low-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.
Low-frequency TDD	Multi-frequency beam coordination extension	**MULTI_FREQ_BEAM_COORD_EXT_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The conditions for triggering the use of virtual grid models in this function are the same as those for triggering this function. Virtual grid models can be used in this function only when the switch controlling this function is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Allow Virtual Grid Model Build Flag	**VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	● Grid-based handover protection (controlled by the **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr**.***HoProtectAlgoSwitch*** parameter) can take effect for a cell only if the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter in at least one **NRCellFreqRelation** MO of this cell is selected. ● Association of NR coverage-type virtual grid models with TA characteristics (controlled by the **NR_COV_VG_MDL_WITH_TA_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can take effect only when both the

RAT	Function Name	Function Switch	Reference	Description
				VG_MODEL_ALLOW_BUILD_FLAG option and the **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter are selected. ● The **CARR_ADD_MEAS_QTY_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter and the **COV_IF_HO_MEAS_QTY_PRED_SW**, **SERV_IF_HO_MEAS_QTY_PRED_SW**, **ULINTRF_IF_HO_MEAS_QTY_PRED_SW**, and **FRQ_PRI_IF_HO_MEAS_QTY_PRED_SW** options of the **NRCellAlgoSwitch**.***AiInterFreqHoOptSwitch*** parameter can be selected for a cell only if the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRel**

RAT	Function Name	Function Switch	Reference	Description
				ation.***AggregationAttribute*** parameter in at least one **NRCellFreqRelation** MO of this cell is selected. ● Measurement quantity prediction for intra-NR inter-frequency handovers (controlled by the **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can take effect only when the **VG_MODEL_ALLOW_BUILD_FLAG** option is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Measurement quantity prediction for intra-NR inter-frequency handovers	**NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	Association of NR coverage-type virtual grid models with TA characteristics (controlled by the **NR_COV_VG_MDL_WITH_TA_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can take effect only when both the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter and the **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option are selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	● Intra-frequency spectral efficiency prediction accuracy increase (controlled by the **INTRAF_SPCT_EFF_PRED_ACCU_IMP** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can be enabled only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected. ● Association of downlink intra-frequency spectral efficiency grid models with the characteristics of multiple SSB beams (controlled by the **DL_INTRA_EFF_MDL_MULTI_SSB_SW** option of the **NRCellSmartMultiCarr**.***NrM***

RAT	Function Name	Function Switch	Reference	Description
				ultiCarrierAlgoSwitch parameter) can take effect only when both the **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter and the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option are selected. ● Downlink intra-frequency spectral efficiency prediction (controlled by the **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) takes effect only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	Intra-frequency spectral efficiency prediction accuracy increase (controlled by the **INTRAF_SPCT_EFF_PRED_ACCU_IMP** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can be enabled only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-frequency spectral efficiency prediction	**DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	Association of downlink intra-frequency spectral efficiency grid models with the characteristics of multiple SSB beams (controlled by the **DL_INTRA_EFF_MDL_MULTI_SSB_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter) can take effect only when both the **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option and the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter are selected.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*
Low-frequency TDD	Inter-Freq Periodic MR Type	**NRCellFreqRelation**.***InterFreqPrdcMrType*** set to **PERIODIC_MR_ONLY**	*SA Mobility Management in Connected Mode*

4.3.2.3 Function Impacts

- Impact relationships with functions related to power saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	**INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After intelligent carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	**LNR_SMART_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*	After LTE and NR intelligent carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When RF channel intelligent shutdown and Virtual Grid-based Multi-Frequency Coordination are both enabled: ● If RF channel intelligent shutdown is not in effect in a local cell and is in effect in its neighboring cells, building of virtual grid models in the local cell may be suspended, and existing virtual grid models of the local cell may go offline and become unavailable. To increase the proportion of online virtual grid models in this scenario, optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions can be enabled on the local and neighboring base stations by selecting the **VIRTUAL_GRID_CH_SD_COOP_OPT_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter. The optimization

RAT	Function Name	Function Switch	Reference	Description
				reduces the probability that building of virtual grid models in the local cell is suspended and the probability that existing virtual grid models of the local cell go offline and become unavailable. ● If RF channel intelligent shutdown is in effect in a local cell, building of virtual grid models in the local cell will be suspended and existing virtual grid models of the local cell will go offline and become unavailable.
Low-frequency TDD	pRRU deep dormancy in scheduled mode	**RRU**.***DORMANCYSW***	*Multi-RAT Coordinated Energy Saving*	After pRRU deep dormancy takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	**TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After TTI-level channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable. If virtual grid models need to be used while TTI-level channel shutdown is in effect, the **VIRTUAL_GRID_CH_SD_COEXIST_SW** option of the **gNBPwrSavingAlgo**.***PwrSavingAlgoSwitch*** parameter must be selected for both the local and neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	**RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*	After dynamic RF channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable. If virtual grid models need to be used while dynamic RF channel shutdown is in effect, the **VIRTUAL_GRID_CH_SD_COEXIST_SW** option of the **gNBPwrSavingAlgo**.***PwrSavingAlgoSwitch*** parameter must be selected for both the local and neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: • **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (intra-base-station multi-carrier scenarios) • **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (inter-base-station multi-carrier scenarios) • **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter Basic-layer cells: • **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*	After fast carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	**FAST_DEEP_DORM_SW** option of the **RRUEXTINFO**.***RRUDOROPTFEATSW*** parameter	*Multi-RAT Coordinated Energy Saving*	After RF module fast deep dormancy takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter	*ANR*	After intra-RAT ANR is enabled, neighbor relationships are automatically added and removed. The base station periodically monitors the changes in the proportions of reported intra-frequency and inter-frequency neighboring cells to determine whether to rebuild a virtual grid model.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	**RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	• If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is deselected and remote interference adaptive avoidance is in effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable. • If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected for both the local and neighboring base stations and remote interference adaptive avoidance is in effect, virtual grid models can be used in the following scenarios: – The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRim**

RAT	Function Name	Function Switch	Reference	Description
				Config.***RimPolicySwitch*** parameter is selected. – The maximum number of SSB beams remains unchanged.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced remote interference adaptive avoidance	**RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	● If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is deselected and enhanced remote interference adaptive avoidance is in effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and become unavailable. ● If the **NR_VGRID_AND_RIM_AVD_COLLAB_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected for both the local and neighboring base stations and enhanced remote interference adaptive avoidance is in effect, virtual grid models can be used in the following scenarios: – The **FORCED_RIM_AVOID_SW** option of the **NRDUCellRim**

RAT	Function Name	Function Switch	Reference	Description
				Config.***RimPolicySwitch*** parameter is selected. – The maximum number of SSB beams remains unchanged.
Low-frequency TDD	Non-gap-assisted measurements for active BWP	**SSB_MEAS_WITHOUT_GAP_SW** option of the **NRDUCellMeas**.***SsbMeasPolicySwitch*** parameter	*SA Mobility Management in Connected Mode*	When non-gap-assisted measurements for active BWP and Virtual Grid-based Multi-Frequency Coordination are both enabled, the latter function preferentially takes effect. That is, virtual grid-based prediction is preferentially performed to select the optimal inter-frequency cell for an inter-frequency handover. If virtual grid-based prediction fails, whether to perform non-gap-assisted measurements for active BWP is determined when measurement configurations are delivered.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Carrier aggregation	CA works in multiple scenarios. Its function switch varies depending on scenarios. For details, see *Carrier Aggregation*.	*Carrier Aggregation*	If the **NR_TO_NR_COV_GRID_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected, data distribution to SCells is delayed. Therefore, the average downlink throughput of CA UEs, which is equal to (**N.CA.SccActiveUser.ThpVol.DL** – **N.CA.ThpVol.DL.LastSlot**)/**N.CA.ThpTime.DL.RmvLastSlot**, slightly decreases in burst service scenarios.
Low-frequency TDD	Cell-level MRO	• **MRO_SCENARIO_IDENT_SW** option of the **gNBMro**.***ScenarioIdentifySwitch*** parameter • **INTRA_FREQ_MRO_SW** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When the use of virtual grid models in inter-frequency handovers and cell-level intra-frequency MRO are both enabled, CIO adjustment by MRO affects the occasion for intra-frequency handovers to trigger event A3. As a result, the number of inter-frequency handovers may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-level MRO	**UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	If the use of virtual grid models in inter-frequency handovers and UE-level MRO are both enabled and the use of virtual grid models in inter-frequency handovers takes effect for ping-pong UEs, CIO adjustment by MRO affects the occasion for intra-frequency handovers to trigger event A3. As a result, the number of inter-frequency handovers may be affected.
Low-frequency TDD	Multiple BWP1s for RedCap	**NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** set to a value other than **20M**	*RedCap*	During virtual grid model building and maintenance, data of RedCap UEs using BWP1s with NCD-SSBs is not collected, and virtual grid models are not used for these UEs, either.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The main control board must be a UMPTg, UMPTga, UMPTe, UMPTha, or UMPThb. There are no requirements for BBPs.

NOTE

UMPTe series main control boards working in co-MPT multiple modes do not support the building of virtual grid models.

- RSRP prediction models, downlink intra-frequency spectral efficiency prediction models, and intra-frequency SRS beam prediction models

  Table 4-10 lists the virtual grid model specifications supported by each type of main control board working in different modes. The number of virtual grid models of a gNodeB equals the total number of virtual grid models of all cells served by the gNodeB.

  For RSRP prediction models, when the **NR_TO_NR_COV_GRID_SW** option of the **gNodeBAlgo**.***NrSmartMultiCarrSw*** parameter is selected, model building is simplified and the maximum allowed number of models is increased.

Table 4-10 Specifications of virtual grid models

Main Control Board	Working Mode	RSRP Prediction Model Specifications		Downlink Intra-Frequency Spectral Efficiency Prediction Model Specifications	Intra-Frequency SRS Beam Prediction Model Specifications
		NR_TO_NR_COV_GRID_SW Deselected	NR_TO_NR_COV_GRID_SW Selected		
UMPTg	NR-only	36	81	30	30
	Co-MPT multiple modes	36	81	30	30
UMPTga	NR-only	36	54	30	30
	Co-MPT multiple modes	18	36	15	15
UMPTe	NR-only	36	54	30	30
	Co-MPT multiple modes	0	0	0	0

Main Control Board	Working Mode	RSRP Prediction Model Specifications		Downlink Intra-Frequency Spectral Efficiency Prediction Model Specifications	Intra-Frequency SRS Beam Prediction Model Specifications
		NR_TO_NR_COV_GRID_SW Deselected	NR_TO_NR_COV_GRID_SW Selected		
UMPTha	NR-only	36	81	30	30
	Co-MPT multiple modes	36	81	30	30
UMPThb	NR-only	36	81	30	30
	Co-MPT multiple modes	36	81	30	30

- Downlink-frequency spectral efficiency models

  Table 4-11 lists the downlink-frequency spectral efficiency model specifications supported by each type of main control board working in different modes. The number of virtual grid models of a gNodeB equals the total number of virtual grid models of all frequencies of the gNodeB.

Table 4-11 Specifications of downlink-frequency spectral efficiency models

Main Control Board	Working Mode	Downlink-Frequency Spectral Efficiency Model Specifications
UMPTg	NR-only	8
	Co-MPT multiple modes	8
UMPTga	NR-only	8
	Co-MPT multiple modes	4
UMPTe	NR-only	4
	Co-MPT multiple modes	0

Main Control Board	Working Mode	Downlink-Frequency Spectral Efficiency Model Specifications
UMPTha	NR-only	8
	Co-MPT multiple modes	8
UMPThb	NR-only	8
	Co-MPT multiple modes	8

NOTE

For a single frequency of a base station, a maximum of 20 cells operating on this frequency can be engaged in the building of a downlink-frequency spectral efficiency model.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

Multi-frequency networking is used.

4.3.5 Others

In NSA networking, Xn interfaces must be configured between gNodeBs.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-12 and **Table 4-13** describe the parameters used for function activation and optimization, respectively.

Table 4-12 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	Aggregation Attribute	**NRCellFreqRelation.*AggregationAttribute***	VG_MODEL_ALLOW_BUILD_FLAG	Select this option if RSRP prediction model building is required based on the network plan.
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr.*NrMultiCarrierAlgoSwitch***	NR_IF_HO_MEAS_QTY_PRED_FUN_SW	Select this option if RSRP prediction model building is required based on the network plan.
Low-frequency TDD	Model Building Switch	**NRCellSmartMulti-Carr**.***ModelBuildingSwitch***	DL_INTRAF_SE_MODEL_BUILD_ALLOW	Select this option if downlink intra-frequency spectral efficiency prediction model building is required based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr.*NrMultiCarrierAlgoSwitch***	DL_INTRA_FREQ_SPCT_EFF_PRED_SW	Select this option if downlink intra-frequency spectral efficiency prediction model building is required based on the network plan.
Low-frequency TDD	Model Building Switch	**NRCellSmartMulti-Carr.*ModelBuildingSwitch***	INTRAF_SRS_MODEL_BUILD_ALLOW	Select this option if intra-frequency SRS beam prediction model building is required based on the network plan.
Low-frequency TDD	Model Building Switch	**NRCellSmartMulti-Carr.*ModelBuildingSwitch***	DL_FREQ_SE_MODEL_BUILD_ALLOW	Select this option if downlink-frequency spectral efficiency model building is required based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	AI-based Inter-Frequency Handover Opt Sw	**NRCellAlgoSwitch.*AiInterFreqHoOptSwitch***	COV_IF_HO_MEAS_QTY_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for coverage-based inter-frequency handovers based on the network plan.
Low-frequency TDD	AI-based Inter-Frequency Handover Opt Sw	**NRCellAlgoSwitch.*AiInterFreqHoOptSwitch***	SERV_IF_HO_MEAS_QTY_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for service-based inter-frequency handovers based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	AI-based Inter-Frequency Handover Opt Sw	**NRCellAlgoSwitch.*AiInterFreqHoOptSwitch***	ULINTRF_IF_HO_MEAS_QTY_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for uplink-interference-based inter-frequency handovers based on the network plan.
Low-frequency TDD	AI-based Inter-Frequency Handover Opt Sw	**NRCellAlgoSwitch.*AiInterFreqHoOptSwitch***	FRQ_PRI_IF_HO_MEAS_QTY_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for frequency-priority-based inter-frequency handovers based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MF_BEAM_CO_EXT_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for multi-frequency beam coordination based on the network plan.
Low-frequency TDD	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	CARR_ADD_MEAS_QTY_PRED_SW	Select this option if the prediction results of RSRP prediction models need to be used for SCell configuration for CA or SUL addition based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Smart Multi-Carrier Switch	**gNodeBAlgo**.***NrSmartMultiCarrSw***	NR_VGRID_AND_RIM_AVD_COLLAB_SW	Select this option if remote interference adaptive avoidance or enhanced remote interference adaptive avoidance needs to take effect together with Virtual Grid-based Multi-Frequency Coordination based on the network plan.
Low-frequency TDD	Power Saving Algorithm Switch	**gNBPwrSavingAlgo**.***PwrSavingAlgoSwitch***	VIRTUAL_GRID_CH_SD_COEXIST_SW	Select this option if TTI-level channel shutdown or dynamic RF channel shutdown needs to take effect together with Virtual Grid-based Multi-Frequency Coordination based on the network plan.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Smart Multi-Carrier Switch	**gNodeBAlgo**.***NrSmartMultiCarrSw***	VIRTUAL_GRID_CH_SD_COOP_OPT_SW	Select this option based on the network plan if the probability of suspending the building of virtual grid models and the probability of virtual grid models going offline and becoming unavailable need to be reduced while RF channel intelligent shutdown is in effect.

Table 4-13 Parameters used for optimization

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr**.***NrMultiCarrierAlgoSwitch***	NR_VG_MDL_MR_IMP_SW	It is recommended that this option be selected to improve the measurement accuracy of virtual grid models.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr.*NrMultiCarrierAlgoSwitch***	NR_COV_VG_MDL_WITH_TA_SW	It is recommended that this option be selected to improve the accuracy of virtual grid models.
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr.*NrMultiCarrierAlgoSwitch***	INTRAF_SPCT_EFF_PRED_ACCU_IMP	Select this option.
Low-frequency TDD	NR Multi-Carrier Algorithm Switch	**NRCellSmartMulti-Carr.*NrMultiCarrierAlgoSwitch***	DL_INTRA_EFF_MDL_MULTI_SSB_SW	It is recommended that this option be selected to improve the accuracy of virtual grid models.
Low-frequency TDD	Prediction Accuracy for Data Thld	**NRCellSmartMulti-Carr.*PredAccuForDataThld***	None	Retain the default value.
Low-frequency TDD	Prediction Accuracy for Voice Thld	**NRCellSmartMulti-Carr.*PredAccuForVoiceThld***	None	Retain the default value.
Low-frequency TDD	NR IF HO Success Rate Protect Thld	**NRCellSmartMulti-Carr.*NrIfHoSuccRateProtectThld***	None	Retain the default value.
Low-frequency TDD	Handover Protect Algorithm Switch	**NRCellSmartMulti-Carr.*HoProtectAlgoSwitch***	MEAS_QTY_PRED_HO_FORBID_SW	Retain the default value.
Low-frequency TDD	Coverage Grid Model Apply Policy	**NRCellSmartMulti-Carr.*CovGridModelAppPolicy***	None	Retain the default value.

RAT	Parameter Name	Parameter ID	Option	Setting Notes
Low-frequency TDD	NR Smart Multi-Carrier Switch	**gNodeBAlgo**.***NrSmartMultiCarrSw***	NR_TO_NR_COV_GRID_SW	Set this option based on operators' requirements.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the RSRP prediction model building switch (low-frequency TDD)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=7853,
AggregationAttribute=VG_MODEL_ALLOW_BUILD_FLAG-1;
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=NR_IF_HO_MEAS_QTY_PRED_FUN_SW-1;
//Turning on the downlink intra-frequency spectral efficiency prediction model building switch
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=DL_INTRA_FREQ_SPCT_EFF_PRED_SW-1,
ModelBuildingSwitch=DL_INTRAF_SE_MODEL_BUILD_ALLOW-1;
//Turning on the intra-frequency SRS beam prediction model building switch (supported only in low-
frequency TDD scenarios)
MOD NRCELLSMARTMULTICARR: NrCellId=0, ModelBuildingSwitch=INTRAF_SRS_MODEL_BUILD_ALLOW-1;
//Turning on the downlink-frequency spectral efficiency model building switch
MOD NRCELLSMARTMULTICARR: NrCellId=0, ModelBuildingSwitch=DL_FREQ_SE_MODEL_BUILD_ALLOW-1;
//Enabling AI-based inter-frequency handover optimization
MOD NRCELLALGOSWITCH: NrCellId=0,
AiInterFreqHoOptSwitch=COV_IF_HO_MEAS_QTY_PRED_SW-1&SERV_IF_HO_MEAS_QTY_PRED_SW-1&ULINT
RF_IF_HO_MEAS_QTY_PRED_SW-1&FRQ_PRI_IF_HO_MEAS_QTY_PRED_SW-1;
//Turning on the multi-frequency algorithm switch
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MF_BEAM_CO_EXT_PRED_SW-1;
//Enabling measurement quantity prediction for carrier addition
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=CARR_ADD_MEAS_QTY_PRED_SW-1;
//Enabling collaboration between NR virtual grid models and remote interference avoidance for the local
base station and all its neighboring base stations when remote interference adaptive avoidance or
enhanced remote interference adaptive avoidance is in effect
MOD GNODEBALGO: NrSmartMultiCarrSw=NR_VGRID_AND_RIM_AVD_COLLAB_SW-1;
//Turning on the switch controlling collaboration between virtual grid models and energy-saving-oriented
channel shutdown functions on the local base station and all neighboring base stations when TTI-level
channel shutdown or dynamic RF channel shutdown is in effect
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=VIRTUAL_GRID_CH_SD_COEXIST_SW-1;
//Turning on the switch controlling optimization of collaboration between virtual grid models and energy-
saving-oriented channel shutdown functions when RF channel intelligent shutdown is in effect
MOD GNODEBALGO: NrSmartMultiCarrSw=VIRTUAL_GRID_CH_SD_COOP_OPT_SW-1;
```

Optimization Command Examples

```
//Turning on the NR multi-carrier algorithm switch
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=NR_VG_MDL_MR_IMP_SW-1&NR_COV_VG_MDL_WITH_TA_SW-1&INTRAF_SPCT_E
FF_PRED_ACCU_IMP-1&DL_INTRA_EFF_MDL_MULTI_SSB_SW-1;
//Turning on the NR-NR coverage grid switch
MOD GNODEBALGO: NrSmartMultiCarrSw=NR_TO_NR_COV_GRID_SW-1;
//Setting parameters related to virtual grid models
MOD NRCELLSMARTMULTICARR: NrCellId=0, NrIfHoSuccRateProtectThld=950,
PredAccuForDataThld=ACCURACY90, PredAccuForVoiceThld=ACCURACY95,
HoProtectAlgoSwitch=MEAS_QTY_PRED_HO_FORBID_SW-1, CovGridModelAppPolicy=FAST_MODE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the multi-frequency algorithm switch
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MF_BEAM_CO_EXT_PRED_SW-0;
//Disabling AI-based inter-frequency handover optimization
MOD NRCELLALGOSWITCH: NrCellId=0,
AiInterFreqHoOptSwitch=COV_IF_HO_MEAS_QTY_PRED_SW-0&SERV_IF_HO_MEAS_QTY_PRED_SW-0&ULINT
RF_IF_HO_MEAS_QTY_PRED_SW-0&FRQ_PRI_IF_HO_MEAS_QTY_PRED_SW-0;
//Disabling measurement quantity prediction for carrier addition
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=CARR_ADD_MEAS_QTY_PRED_SW-0;
//Turning off the NR multi-carrier algorithm switch
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=NR_VG_MDL_MR_IMP_SW-0&NR_COV_VG_MDL_WITH_TA_SW-0&INTRAF_SPCT_E
FF_PRED_ACCU_IMP-0&DL_INTRA_EFF_MDL_MULTI_SSB_SW-0;
//Turning off the NR-NR coverage grid switch
MOD GNODEBALGO: NrSmartMultiCarrSw=NR_TO_NR_COV_GRID_SW-0;
//Disabling collaboration between NR virtual grid models and remote interference avoidance for the local
base station and all its neighboring base stations
MOD GNODEBALGO: NrSmartMultiCarrSw=NR_VGRID_AND_RIM_AVD_COLLAB_SW-0;
//Turning off the switch controlling collaboration between virtual grid models and energy-saving-oriented
channel shutdown functions on the local base station and all neighboring base stations
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=VIRTUAL_GRID_CH_SD_COEXIST_SW-0;
//Turning off the switch controlling optimization of collaboration between virtual grid models and energy-
saving-oriented channel shutdown functions
MOD GNODEBALGO: NrSmartMultiCarrSw=VIRTUAL_GRID_CH_SD_COOP_OPT_SW-0;
//Turning off the RSRP prediction model building switch (low-frequency TDD)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=7853,
AggregationAttribute=VG_MODEL_ALLOW_BUILD_FLAG-0;
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=NR_IF_HO_MEAS_QTY_PRED_FUN_SW-0;
//Turning off the downlink intra-frequency spectral efficiency prediction model building switch
MOD NRCELLSMARTMULTICARR: NrCellId=0,
NrMultiCarrierAlgoSwitch=DL_INTRA_FREQ_SPCT_EFF_PRED_SW-0,
ModelBuildingSwitch=DL_INTRAF_SE_MODEL_BUILD_ALLOW-0;
//Turning off the intra-frequency SRS beam prediction model building switch (supported only in low-
frequency TDD scenarios)
MOD NRCELLSMARTMULTICARR: NrCellId=0, ModelBuildingSwitch=INTRAF_SRS_MODEL_BUILD_ALLOW-0;
//Turning off the downlink-frequency spectral efficiency model building switch
MOD NRCELLSMARTMULTICARR: NrCellId=0, ModelBuildingSwitch=DL_FREQ_SE_MODEL_BUILD_ALLOW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Observation of Virtual Grid Model Building

Sufficient data must be collected for virtual grid model building. Therefore, you are advised to wait for at least 24 hours after starting virtual grid model building and then use one of the following methods to observe the model building status:

- Using MML commands

  Run the **DSP NRCELLAIGRIDMODEL** command to query virtual grid model information.

  – If the value of **RSRP Forecast Status** in the NR-NR virtual grid model information is **Available**, the RSRP prediction model has been successfully built.
  – If the value of **Ai Grid Model Status** in the downlink intra-frequency spectral efficiency grid model information is **Available**, the downlink intra-frequency spectral efficiency prediction model has been successfully built.
  – If the value of **Ai Grid Model Status** in the intra-frequency SRS beam grid model information is **Available**, the intra-frequency SRS beam prediction model has been successfully built.
  – If the value of **Ai Grid Model Status** in the downlink-frequency spectral efficiency grid model information is **Available**, the downlink-frequency spectral efficiency model has been successfully built.
- Using counters
  – RSRP prediction models can be observed using the **N.NrVirtualGrid.Model.Avg** and **N.NrVirtualGrid.Model.Max** counters. If the value of either counter is not 0, one or more RSRP prediction models have been successfully built.
  – Downlink intra-frequency spectral efficiency prediction models can be observed using the **N.NrDlSEGrid.Model.Avg** counter. If the value of this counter is not 0, one or more downlink intra-frequency spectral efficiency prediction models have been successfully built.

It takes a week to fully build a virtual grid model. Normally, a virtual grid model does not come online within a week after the function is enabled. If a virtual grid model is still not online after a week, perform the following operations:

1. Run the **DSP NRCELLAIGRIDMODEL** command. If the value of **RSRP Model Latest Offline Reason** in the NR-NR virtual grid model information is **N/A**, the RSRP prediction model has not come online. If the value of **Ai Grid Model Latest Offline Reason** in the downlink intra-frequency spectral efficiency grid model information is **N/A**, the downlink intra-frequency spectral efficiency prediction model has not come online. If the value of **Ai Grid Model Latest Offline Reason** in the intra-frequency SRS beam grid model information is **N/A**, the intra-frequency SRS beam prediction model has not come online. If the value of **Ai Grid Model Latest Offline Reason** in the downlink-frequency spectral efficiency grid model information is **N/A**, the downlink-frequency spectral efficiency model has not come online. If the displayed value of any of the preceding parameters is not **N/A**, the value indicates the reason why the model is not online, and the time when the model went offline is indicated by the corresponding parameter in the command output.

2. If the RSRP predication model, intra-frequency SRS beam prediction model, or downlink intra-frequency spectral efficiency prediction model has not come online, collect the average value of **N.User.RRCConn.Avg** over the last seven days. An average value less than five indicates that the model has not come online because of insufficient samples. If this is the case, no further locating is required. If the average value is greater than or equal to five, contact Huawei engineers for fault locating. If the downlink-frequency spectral efficiency model has not come online, collect the average value of **N.User.RRCConn.Avg** over the last seven days. An average value less than two indicates that the model has not come online because of insufficient samples. If this is the case, no further locating is required. If the average value is greater than or equal to two, the number of intra-frequency cells on the base station is greater than or equal to three, and the downlink-frequency model switch is on for all the cells, contact Huawei engineers for fault locating.

NOTE

- When the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellFreqRelation**.***AggregationAttribute*** parameter is selected and the **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is deselected, information about the RSRP prediction model for the corresponding frequency can be queried by running the **DSP NRCELLAIGRIDMODEL** command. However, the model is not built and occupies a unit in the number of models.
- When the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected and the **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter is deselected, information about the downlink intra-frequency spectral efficiency prediction model for the corresponding frequency can be queried by running the **DSP NRCELLAIGRIDMODEL** command. However, the model is not built and occupies a unit in the number of models.

Observation of the Use of Virtual Grid Models

Observe the following counters listed in **Table 4-14**. If the value of any counter is greater than 0, RSRP prediction models have been used.

Table 4-14 Performance counters related to the use of RSRP prediction models

Counter ID	Counter Name
1911831557	**N.HO.NrVirtualGrid.Model.ExecAttOut**
1911831556	**N.HO.NrVirtualGrid.Model.ExecSuccOut**
1911831558	**N.HO.NrVirtualGrid.Model.PrepAttOut**
1911831564	**N.CA.SCell.VirtualGrid.Add.Succ**
1911831565	**N.CA.SCell.VirtualGrid.Add.Att**
1911831567	**N.NrVirtualGrid.Model.InterFreq.MeasFree.Times**
1911831619	**N.NsaDc.NrVirtualGrid.Model.PSCell.Change.Att**
1911831620	**N.NsaDc.NrVirtualGrid.Model.PSCell.Change.Succ**

4.4.3 Network Monitoring

Observe the value of **User Downlink Average Throughput (DU)** to check the increase in UE throughput after this feature is enabled.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *ANR*
- *Carrier Aggregation*
- *High Speed Mobility*
- *Cell Combination*
- *SA Mobility Management in Connected Mode*
- *Hyper Cell*
- *Remote Interference Management (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *Multi-Frequency Smart Selection*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*
- *MRO*
- *RedCap*
- *License Control Item Lists*
- *Super Uplink*